

MSDI Calculator

Weather Data Methodology and Metadata

Technical Methodology Memorandum

Prepared for internal model documentation and technical review

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Model scope: Manure Sub-Surface Drip Irrigation (mSDI-e) economic screening calculator

Document status: *Methodology documentation for weather data acquisition and derivatives creation used in current and future app versions.*

Purpose and Scope

This document describes the weather and climate data methodology used to support baseline and future-scenario water demand calculations in the MSDI Calculator. The objective is to provide a defensible, reproducible technical record for how historical seasonal evapotranspiration and precipitation data are prepared, how future climate scenario data are converted into agricultural reference evapotranspiration, and how county-level lookup values are selected for model use.

The methodology is written for technical documentation, model review, and future implementation. It does not implement the scenario logic in the application by itself. It documents the intended weather-data processing workflow, metadata standards, assumptions, and quality-control checks that should accompany the prepared lookup datasets.

Terminology and Units

The model distinguishes among reference evapotranspiration, potential evapotranspiration, precipitation, and crop water demand. This distinction is important because the historical baseline and future climate projection products do not all represent the same evapotranspiration concept.

Term	Meaning in this workflow	Use in MSDI Calculator
ET0 / ETr	Alfalfa-reference evapotranspiration, as used in the project workflow for agricultural reference demand. Public ClimateEngine gridMET documentation lists ASCE Alfalfa Reference Evapotranspiration as a gridded variable.	Primary baseline evapotranspiration input for irrigation demand calculations.
PET	Potential evapotranspiration from EPA EnviroAtlas climate scenario datasets. PET is a generalized atmospheric demand variable and is not treated as equivalent to alfalfa-reference ET.	Intermediate projection variable converted to ET0 using seasonal adjustment factors.
Precipitation	Seasonal total precipitation for baseline and future scenario periods.	Used to account for climate water inputs alongside ET0.
Season	DJF, MAM, JJA, and SON seasonal periods.	All baseline and scenario values are processed seasonally.

Reference ET terminology varies by agency and region. USGS distinguishes reference ET variants for short grass and tall alfalfa reference crops, while Reclamation AgriMet documentation describes the use of alfalfa as the reference crop for crop water-use charts. The MSDI Calculator workflow uses the alfalfa-reference ET concept because it is more directly aligned with crop water demand modeling than PET. [4, 5]

Part 1. Source Datasets

1.1 Historical Baseline ET0 and Precipitation

Historical baseline seasonal ET0 and precipitation values were downloaded from ClimateEngine for the 1991-2020 climate-normal period. The project workflow refers to the ET0 source as AgriMet/alfalfa-reference ET accessed through ClimateEngine. For final publication of any dataset-specific metadata, the source name in the exported raster metadata should be retained exactly. Public ClimateEngine documentation for gridMET identifies a 4-km, daily, CONUS dataset beginning in 1979 and lists ASCE Alfalfa Reference Evapotranspiration and precipitation variables. [1]

gridMET is a gridded surface meteorological dataset that blends PRISM spatial climatology with NLDAS-2 temporal meteorology using climatically aided interpolation. This makes it suitable for regional agricultural and hydrologic modeling where spatially complete daily weather inputs are required. [2, 3]

Baseline variable	Source used in workflow	Temporal period	Seasonal processing	Primary reason for use
Alfalfa-reference ET0	ClimateEngine download; project workflow labels it as AgriMet/alfalfa-reference ET	1991-2020	DJF, MAM, JJA, SON seasonal means or totals as appropriate to the exported raster metric	Agricultural reference evapotranspiration suitable for irrigation demand modeling.
Precipitation	ClimateEngine precipitation dataset associated with the same baseline workflow	1991-2020	DJF, MAM, JJA, SON seasonal precipitation totals	Historical seasonal water input for baseline calculations.

1.2 EPA EnviroAtlas Historical PET and Future Scenario Data

EPA EnviroAtlas climate scenario data were used for historical PET and future PET and precipitation projections. EnviroAtlas climate scenario tools include four RCP scenarios - RCP 2.6, RCP 4.5, RCP 6.0, and RCP 8.5 - and climate variables including precipitation and evapotranspiration. [6]

The EnviroAtlas PET metadata describe the dataset as a 1950-2099 product for the conterminous United States generated from NASA Earth Exchange NEX-DCP30 downscaled climate projections using ensemble averages of more than 30 available climate models for each RCP at approximately 30 arc-second spatial resolution. [7]

The EnviroAtlas precipitation fact sheet documents projected seasonal and annual precipitation for the contiguous United States using four future greenhouse-gas concentration scenarios. In this workflow, seasonal precipitation projections are retained directly as precipitation scenario inputs; PET projections are adjusted to ET0 as described below. [8]

Scenario product	Variables used	RCPs	Time points	Seasonal processing
EnviroAtlas historical climate scenarios	Historical PET	Historical ensemble baseline	1991-2020 processing period for adjustment-factor derivation	DJF, MAM, JJA, SON
EnviroAtlas future climate scenarios	Future PET and precipitation	RCP 2.6, RCP 4.5, RCP 6.0, RCP 8.5	2030, 2050, 2080	DJF, MAM, JJA, SON

Part 2. ETO Projection Methodology

2.1 Why PET Is Not Used Directly

PET and alfalfa-reference ET0 are not interchangeable. PET is a generalized potential water-loss metric, while alfalfa-reference ET represents evapotranspiration from a standardized tall reference crop and is more suitable for agricultural crop water demand calculations. Using PET directly as ET0 would risk introducing systematic bias in crop-water-demand estimates, particularly when the model uses crop coefficients or irrigation demand relationships calibrated to reference ET.

For that reason, EnviroAtlas future PET projections are used only as climate-change projection surfaces. They are converted to an estimated agricultural reference ET0 surface using season-specific spatial adjustment factors derived from the relationship between historical ET0 and historical PET.

2.2 Seasonal PET-to-ET0 Adjustment Factors

For each season, a spatial adjustment-factor raster is calculated by dividing the historical alfalfa-reference ET0 raster by the corresponding historical PET raster. This produces four seasonal adjustment-factor rasters: DJF, MAM, JJA, and SON.

$$\text{Equation 1. Seasonal adjustment factor} = \text{HistoricalET0_season} / \text{HistoricalPET_season}$$

The adjustment factor is dimensionless. It is calculated cell-by-cell after rasters have been aligned to a common spatial grid or otherwise processed to ensure spatial compatibility. Cells with missing, zero, or invalid PET values must be masked or handled according to the quality-control rules in Section 7.

2.3 Future ET0 Projection Rasters

Future ET0 projection rasters are generated by multiplying each future seasonal EnviroAtlas PET projection raster by the corresponding seasonal adjustment-factor raster.

$$\text{Equation 2. ProjectedET0_RCP,time,season} = \text{FuturePET_RCP,time,season} \times \text{AdjustmentFactor_season}$$

This approach preserves the spatial and scenario structure of the EnviroAtlas PET projections while anchoring the magnitude of projected evapotranspiration to the historical agricultural ET0 baseline used by the MSDI Calculator.

Precipitation Projection Methodology

Historical baseline precipitation and EnviroAtlas future precipitation projections are treated as precipitation variables and summarized seasonally. Baseline precipitation is processed for 1991-2020, and future precipitation (planned in version 2) will be processed for each RCP-time-season combination.

The precipitation product is reported using the units of the source rasters or the standardized model lookup units after conversion. If source rasters are in inches and the model lookup requires inches, no unit conversion is needed. If source rasters are in millimeters, values are converted consistently using raster calculator within GIS and audited prior to extracting zonal statistics and lookup-table generation.

Seasonal Definitions

Season code	Season name	Months included	Notes
DJF	Winter	December, January, February	Use a consistent year assignment rule when aggregating December with January-February. Document whether DJF is assigned to the January-February year or the December year.
MAM	Spring	March, April, May	Seasonal aggregation aligned with EnviroAtlas and baseline raster workflow.
JJA	Summer	June, July, August	Seasonal aggregation aligned with EnviroAtlas and baseline raster workflow.
SON	Fall	September, October, November	Seasonal aggregation aligned with EnviroAtlas and baseline raster workflow.

Spatial Aggregation and Lookup Hierarchy

Final model lookup values are generated by spatially summarizing climate rasters by county and by Cropland Data Layer (CDL) land-class grouping. The intended two CDL groupings are cultivated and non-cultivated. Cultivated summaries receive priority because they better represent the land base where irrigation and crop water demand calculations are most relevant.

Priority	Lookup value used by model	Condition
1	County cultivated mean	Use when a valid county cultivated ET0 or precipitation value exists for the selected season/scenario.
2	County non-cultivated mean	Use when county cultivated value is missing but county non-cultivated value exists.
3	State average	Use when county-level cultivated and non-cultivated values are unavailable.
4	National average	Use only when county and state values are unavailable.

The fallback hierarchy should be applied independently for each variable, season, RCP, and time point. For example, a county might use cultivated ET0 for JJA RCP 4.5 2050 but a state-average precipitation value for DJF RCP 8.5 2080 if county precipitation coverage is missing.

Weather Lookup Table Metadata Fields

Each lookup table or generated JSON record should retain enough metadata to trace each value back to its source raster and processing path. At minimum, the prepared lookup should include the fields below.

Field	Description
variable	ET0, PET, precipitation, or adjustment factor.
source_dataset	ClimateEngine/gridMET, AgriMet-labelled ClimateEngine export, EPA EnviroAtlas PET, or EPA EnviroAtlas precipitation.
source_variable	Exact variable name from download metadata.
period_type	baseline, historical_pet, or future_scenario.
baseline_period	1991-2020 where applicable.
rcp	RCP 2.6, 4.5, 6.0, 8.5, or none for baseline.
time_point	2030, 2050, 2080, or none for baseline.
season	DJF, MAM, JJA, SON.
geography_level	county_cultivated, county_non_cultivated, state_average, national_average.
county_fips	County FIPS code where applicable.
state_fips	State FIPS code where applicable.
cdl_group	cultivated, non_cultivated, all, state, or national.
units	Final model units after any conversion.
value	Final numeric lookup value.
source_raster_id	Raster filename or internal asset ID.
processing_date	Date lookup was generated.
fallback_used	Boolean or categorical indicator of whether fallback was used.
fallback_source	Which fallback level supplied the value.

Quality Control and Validation Checks

The following checks should be completed before finalizing lookup tables for model use:

1. Confirm that all rasters use consistent units before raster arithmetic or tabular aggregation.
2. Confirm that historical ET0 and historical PET rasters are spatially aligned before calculating adjustment factors.
3. Mask cells where historical PET is zero, missing, or otherwise invalid before division.
4. Review seasonal adjustment-factor rasters for extreme outliers, especially in arid regions, coastal cells, and cells with sparse valid data.
5. Validate that adjusted future ET0 is within physically plausible seasonal ranges compared with the baseline ET0 distribution.
6. Confirm that precipitation values are not inadvertently converted with evapotranspiration adjustment factors.
7. Check county-level cultivated and non-cultivated summaries for minimum valid cell counts before accepting a mean.
8. Record fallback usage so model outputs can identify where county cultivated data were unavailable.
9. Perform spot checks for representative counties in major dairy regions and compare baseline seasonal values with expected local climate patterns.
10. Archive source raster filenames, processing scripts, and generated lookup tables together to preserve reproducibility.

Assumptions and Limitations

- The PET-to-ET0 adjustment approach assumes the historical spatial relationship between PET and alfalfa-reference ET0 is an appropriate seasonal scaling relationship for future PET projections.
- The method adjusts the magnitude of PET projections but does not explicitly correct future changes in wind, humidity, radiation, or crop-specific physiological response beyond what is represented in the source PET projection and historical adjustment factor.
- RCP scenario outputs should be interpreted as scenario-based planning estimates rather than forecasts of a single expected future.
- County and CDL summaries reduce local spatial variability. Field-level design decisions should use site-specific climate, soil, water, and management information where available.
- The DJF seasonal period crosses calendar years; the selected year-assignment rule should remain consistent across baseline and future processing.
- The current methodology documents planned model integration. Implementation should not proceed until lookup tables have been finalized, tested, and reviewed.

Processing Workflow Summary

1. Download or prepare 1991-2020 seasonal baseline alfalfa-reference ET0 rasters from the ClimateEngine workflow.
2. Download or prepare 1991-2020 seasonal baseline precipitation rasters from the same baseline workflow.
3. Download 1991-2020 historical seasonal PET rasters from EPA EnviroAtlas.
4. For each season, divide historical ET0 by historical PET to create a seasonal adjustment-factor raster.
5. Download future seasonal EnviroAtlas PET and precipitation rasters for each RCP and time point.
6. Multiply each future PET raster by the appropriate seasonal adjustment-factor raster to estimate future ET0.
7. Summarize baseline and future ET0 and precipitation rasters by county and CDL group.
8. Generate county-cultivated, county-non-cultivated, state-average, and national-average lookup records.
9. Apply the fallback hierarchy during model execution or during lookup-table precomputation.
10. Store source metadata, processing date, and fallback information in the lookup output.

References and Source Links

Visible hyperlinks are provided for review and traceability. Accessed May 8, 2026.

[1] ClimateEngine. gridMET Daily 4-km Dataset Documentation.

https://climateengine.org/datasets/climatehydrology/gridmet_daily_4000/

Overview of the ClimateEngine GRIDMET dataset

[2] Climatology Lab. gridMET Overview. <https://www.climatologylab.org/gridmet.html>

Overview of the gridMET high-resolution gridded meteorological dataset

[3] Google Earth Engine Data Catalog. GRIDMET: University of Idaho Gridded Surface Meteorological Dataset.

https://developers.google.com/earth-engine/datasets/catalog/IDAHO_EPSCOR_GRIDMET

List of GRIDMET variables and confirmation of temporal qualities of the dataset

[4] U.S. Bureau of Reclamation. AgriMet: About Crop Water-Use Charts.

https://www.usbr.gov/gp/agrimet/crop_about_charts.html

Description of methods for AgriMet reference ET calculations

[5] U.S. Geological Survey. Evapotranspiration Terminology and Definitions.

<https://www.usgs.gov/publications/evapotranspiration-terminology-and-definitions>

List of reference crop evapotranspiration terminology and definitions

[6] U.S. EPA EnviroAtlas Interactive Map. <https://www.epa.gov/enviroatlas/enviroatlas-interactive-map>

Description of EPA EnviroAtlas climate scenario map tools including RCP 2.6, 4.5, 6.0, and 8.5 and climate variables including precipitation and evapotranspiration.

[7] Data.gov / U.S. EPA. EnviroAtlas - Potential Evapotranspiration 1950-2099 for the Conterminous United States.

<https://catalog.data.gov/dataset/enviroatlas-potential-evapotranspiration-1950-2099-for-the-conterminous-united-states3>

Overview of PET and EnviroAtlas PET climate scenarios derived from NASA Earth Exchange NEX-DCP30

[8] U.S. EPA EnviroAtlas National Data Fact Sheet. Climate Scenarios (1950-2099): Precipitation.

https://enviroatlas.epa.gov/enviroatlas/DataFactSheets/pdf/Supplemental/Climate_Precip.pdf

EPA Enviroatlas fact sheet

[9] U.S. EPA EnviroAtlas Data Download. <https://www.epa.gov/enviroatlas/data-download>

EPA data download page for EnviroAtlas national datasets.

[10] NOAA Climate.gov. Future Climate Projections - Graphs & Maps. <https://www.climate.gov/maps-data/dataset/future-climate-projections-graphs-maps>

General reference for CMIP5 climate-model projections and future scenario concepts.

Appendix A. Future Version Climate Scenarios Functionality

Current App functionality uses 30-year historical climate averages (Seasonal ETo and Precipitation) to feed water estimates within the model such as crop water demand. EPA Enviroatlas provides seasonal rasters for the below RCP pathways and Timepoints. The ability to model against these data will possibly be included in future app versioning.

Scenario	Label	Available time points
RCP 2.6	Low emissions	2030, 2050, 2080
RCP 4.5	Moderate emissions	2030, 2050, 2080
RCP 6.0	Moderate-high emissions	2030, 2050, 2080
RCP 8.5	High emissions	2030, 2050, 2080